

Advanced Python Programming

Level-2 Loop Tracing Practice (No Answers)

Instruction: Predict the output. Clearly show the trace of program execution and justify your answer.

Question 1

```
x=0
for i in range(1,5):
    for j in range(i):
        x+=j
    print(x,end=" ")
```

Question 2

```
x=1
for i in range(1,5):
    x*=i
    if x%3==0: continue
    print(x,end=" ")
```

Question 3

```
for i in range(4):
    for j in range(4):
        if j == 0 and i+j==5: break
    print(i*j,end=" ")
```

Question 4

```
x=0
for i in range(5):
    x^=i
    print(x if x&1 else i,end=" ")
```

Question 5

```
for i in range(1,5):
    for j in range(1,5):
        if i*j>6: break
        if (i+j)%2: continue
    print(i,j,end=" | ")
```

Question 6

```
x=0
for i in range(4):
    for j in range(4):
        if i==j: x+=1
        else: x-=1
print(x)
```

Question 7

```
for i in range(5):
    if i%2:
        for j in range(i):
            if j==1: break
            print(i+j,end=" ")
```

Question 8

```
x=5
for i in range(3):
    x=(x<<1)^i
print(x)
```

Question 9

```
for i in range(3):
    for j in range(3):
        if (i^j)&1:
            print(i+j,end=" ")
        else:
            continue
```

Question 10

```
x=0
for i in range(1,6):
    if i%2==0:
        x+=i
    else:
        x-=i
print(x)
```

Question 11

```
for i in range(4):
    for j in range(4):
        if i==3: break
        if j==2: continue
        print(i+j,end=" ")
```

Question 12

```
x=1
for i in range(1,5):
    for j in range(2):
        x+=i*j
print(x)
```

Question 13

```
for i in range(5,0,-1):
    if i==3: continue
    print(i,end=" ")
```

Question 14

```
x=0
for i in range(4):
    x+=i
    for j in range(i):
        x-=j
print(x)
```

Question 15

```
for i in range(1,5):
    for j in range(i,0,-1):
        if j==2: break
        print(i*j,end=" ")
```

Question 16

```
x=7
for i in range(3):
    x^=(i+1)
print(x)
```

Question 17

```
for i in range(1,5):
    for j in range(1,5):
        if i==j: continue
        if i*j%3==0: break
        print(i+j,end=" ")
```

Question 18

```
x=0
for i in range(3):
    x+=i
    print(x,end=" ")
    x*=2
```

Question 19

```
for i in range(4):
    for j in range(4):
        if (i+j)%3==0: continue
        print(i-j,end=" ")
```

Question 20

```
x=0
for i in range(1,5):
    x+=i
    if x>5: break
print(x)
```